PulseIQ Analytics - Phase 1 Report

Data Analysts Track | Days 1-5 | Prepared 19 October 2026

Name	Poonam Satish Patole	**Phase**	Phase 1
GitHub repository	**PENDING: create private repo and add URL**	**Git tag**	**PENDING: phase-1**

DRAFT - not ready for programme upload. This workspace has no Git repository or GitHub remote yet. The Task 5 screenshot and personal reflection also need to be added/confirmed.

Five-task review

Task	Status	Evidence
1. Load and validate	Done with supplied-contract scope	CSV row-count screenshot; contract notes; pipeline summary. Contracts are supplied for four feeds.
2. Star schema	Partial	11 dimensions and 12 facts built; ERD and orphan audit saved. Only four contracted facts have Task 1 passed-row reconciliation.
3. Quality gatekeeper	Done with findings	14 checks and scratch self-tests; 11/11 planted issue categories caught. Four additional payroll-month differences and freshness noise documented.
4. Governed metrics	Done with limits	28 official views, catalog, privacy suppression and five calculator checks. MAPE has no actuals for forecast months.
5. Workforce dashboard	Built; screenshot pending	Streamlit app has four charts, filters and M29 KPI; runtime check showed no app exceptions.

Learning reflection - personalize these 4 lines before submission

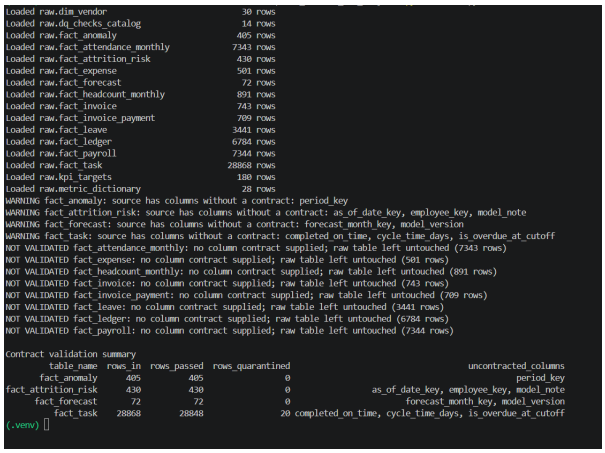
I learned that raw files, validated data, warehouse tables and dashboard metrics should have clear separate roles. A star schema connects facts to dimensions so analysis follows consistent keys and employee history. Data quality checks need known bad examples to prove that they detect problems. A metric definition should be reusable, privacy-aware and traceable to its source.

Problems and how I addressed them

Python commands initially ran from the wrong folder, so the project now uses root-relative commands and a virtual environment. The starter requirements and data were nested under data/, which I confirmed before loading. Task 2 exposed source rows with no matching employee/date dimension version; I recorded their reasons instead of dropping them. Task 3 identified planted errors plus extra payroll and freshness findings, which remain review items.

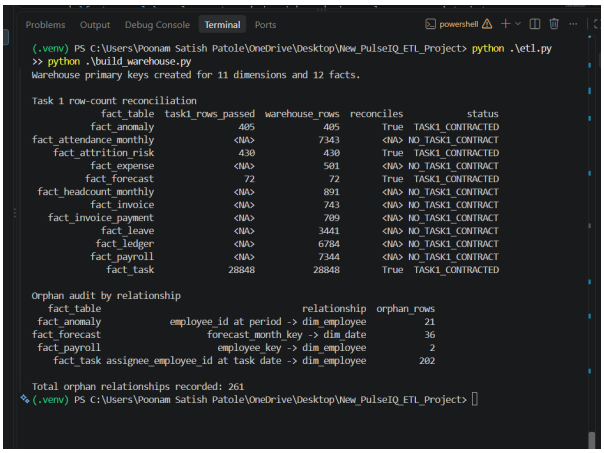
Evidence and checks

Existing screenshots and automated validation results



```
Loaded raw.dim_vendor                          30 rows
Loaded raw.dq_checks_catalog                   14 rows
Loaded raw.fact_anomaly                       405 rows
Loaded raw.fact_attendance_monthly           7343 rows
Loaded raw.fact_attrition_risk                430 rows
Loaded raw.fact_expense                       501 rows
Loaded raw.fact_forecast                       72 rows
Loaded raw.fact_headcount_monthly             891 rows
Loaded raw.fact_invoice                       743 rows
Loaded raw.fact_invoice_payment               709 rows
Loaded raw.fact_leave                        3441 rows
Loaded raw.fact_ledger                       6784 rows
Loaded raw.fact_payroll                      7344 rows
Loaded raw.fact_task                        28868 rows
Loaded raw.kpi_targets                        180 rows
Loaded raw.metric_dictionary                   28 rows
WARNING fact_anomaly: source has columns without a contract: period_key
WARNING fact_attrition_risk: source has columns without a contract: as_of_date_key, employee_key, model_note
WARNING fact_forecast: source has columns without a contract: forecast_month_key, model_version
WARNING fact_task: source has columns without a contract: completed_on_time, cycle_time_days, is_overdue_at_cutoff
NOT VALIDATED fact_attendance_monthly: no column contract supplied; raw table left untouched (7343 rows)
NOT VALIDATED fact_expense: no column contract supplied; raw table left untouched (501 rows)
NOT VALIDATED fact_headcount_monthly: no column contract supplied; raw table left untouched (891 rows)
NOT VALIDATED fact_invoice: no column contract supplied; raw table left untouched (743 rows)
NOT VALIDATED fact_invoice_payment: no column contract supplied; raw table left untouched (709 rows)
NOT VALIDATED fact_leave: no column contract supplied; raw table left untouched (3441 rows)
NOT VALIDATED fact_ledger: no column contract supplied; raw table left untouched (6784 rows)
NOT VALIDATED fact_payroll: no column contract supplied; raw table left untouched (7344 rows)

Contract validation summary
      table_name  rows_in  rows_passed  rows_quarantined                                uncontracted_columns
    fact_anomaly      405          405                 0                                          period_key
fact_attrition_risk   430          430                 0                 as_of_date_key, employee_key, model_note
   fact_forecast       72           72                 0                          forecast_month_key, model_version
       fact_task    28868        28848                20 completed_on_time, cycle_time_days, is_overdue_at_cutoff
(.venv)
```



```
Problems  Output  Debug Console  Terminal  Ports                                   powershell
(.venv) PS C:\Users\Poonam Satish Patole\OneDrive\Desktop\New_PulseIQ_ETL_Project> python .\etl.py
>> python .\build_warehouse.py
Warehouse primary keys created for 11 dimensions and 12 facts.

Task 1 row-count reconciliation
           fact_table  task1_rows_passed  warehouse_rows  reconciles           status
          fact_anomaly                405             405        True  TASK1_CONTRACTED
fact_attendance_monthly               <NA>            7343        <NA> NO_TASK1_CONTRACT
     fact_attrition_risk                430             430        True  TASK1_CONTRACTED
           fact_expense               <NA>             501        <NA> NO_TASK1_CONTRACT
         fact_forecast                 72              72        True  TASK1_CONTRACTED
 fact_headcount_monthly               <NA>             891        <NA> NO_TASK1_CONTRACT
          fact_invoice                <NA>             743        <NA> NO_TASK1_CONTRACT
    fact_invoice_payment              <NA>             709        <NA> NO_TASK1_CONTRACT
             fact_leave               <NA>            3441        <NA> NO_TASK1_CONTRACT
            fact_ledger               <NA>            6784        <NA> NO_TASK1_CONTRACT
           fact_payroll               <NA>            7344        <NA> NO_TASK1_CONTRACT
              fact_task              28848           28848        True  TASK1_CONTRACTED

Orphan audit by relationship
    fact_table                                relationship  orphan_rows
  fact_anomaly          employee_id at period -> dim_employee           21
fact_forecast                 forecast_month_key -> dim_date           36
  fact_payroll                    employee_key -> dim_employee            2
    fact_task assignee_employee_id at task date -> dim_employee          202

Total orphan relationships recorded: 261
(.venv) PS C:\Users\Poonam Satish Patole\OneDrive\Desktop\New_PulseIQ_ETL_Project>
```

Task 1: CSV / raw table count evidence

Task 2: warehouse build and audit evidence

Proof it works

Check	Result
CSV row counts	All 29 CSV tables matched their DuckDB raw-table row counts in the saved check output.
Warehouse build	11 dimensions and 12 facts created. Four contracted fact tables reconcile to Task 1 passed rows; remaining fact tables are labeled NO_TASK1_CONTRACT. 261 relationship exceptions are recorded with explanations.
Data quality	14 checks run; all 14 scratch bad-row self-tests passed. Validation-vault review caught all 11 planted issue categories (0 missed).
Metric layer	28 official metric views created; five values matched source arithmetic. Current MAPE is NULL because forecast months have no actual headcount yet.
Dashboard	Streamlit AppTest rendered 4 charts and 4 KPI cards with no exceptions. Add a browser screenshot to task_5/evidence before final submission.

Traceability and star-schema explanation

A dashboard number can be traced from a metric view to its warehouse fact/dimension inputs and then to raw source tables. For example, headcount is period-end employee count by department and month. A star schema places measurable events or snapshots in fact tables and descriptive context in dimensions. The employee_key points to the employee history version valid for that fact date, so department changes are attributed to the correct period.

Metric challenge and next phase

New metric M29 was defined, added to the metric catalog, shown on the dashboard and checked by hand.

Mini challenge: M29 headcount growth month over month

Definition: (current period-end headcount - previous period-end headcount) / previous period-end headcount x 100. Grain is company-month or department-month. Groups below five are suppressed.

Month	Current	Previous	Hand calculation	View
Sep 2026	441	427	(441 - 427) / 427 x 100 = 3.278689%	+3.28%

The dashboard applies a two-colour meaning consistently: navy for workforce size and teal for workforce composition. It reads from the metrics schema, including `metrics.m29_headcount_growth_mom`.

Task 4 calculator checks

Metric	Hand calculation	Result
M01 Headcount	D01 Sep 2026: period-end headcount	167
M04 Attrition	2 exits / 160 starting headcount x 100	1.25%
M07 Gross pay	D01 Sep 2026 sum across 169 employees	INR 20,720,901.38
M16 Revenue	Account 4001, Apr 2025: negative signed ledger sum	INR 41,824,400
M22 Completion	T01 / SP-01: 50 done / 55 non-cancelled	90.91%

Problems still to resolve

Task 2 passed-row reconciliation is available for four of twelve fact tables because Task 1 has supplied contracts for only four. Task 3 has four additional payroll/ledger month differences and a noisy historical-feed freshness result. Three raw dictionary privacy thresholds are blank; Task 4/5 apply documented effective thresholds. Task 5 browser screenshot, private GitHub repository URL, phase-1 tag and collaborator access have not yet been recorded.

Plan for the next phase

Complete the repository setup and Phase 1 tag, save the dashboard screenshot, and finish the remaining fact reconciliation. Then build the next workforce pages for attendance, leave, attrition and team productivity, resolving quality findings before presenting affected numbers.

Before upload: create the private GitHub repository, add its clickable URL here, create and push the `phase-1` tag, add `fobesitech` as collaborator, capture the current Task 5 page to `task_5/evidence/workforce_dashboard.png`, and rewrite the learning reflection in your own words. Rename this file without `_DRAFT` only after those items are done.